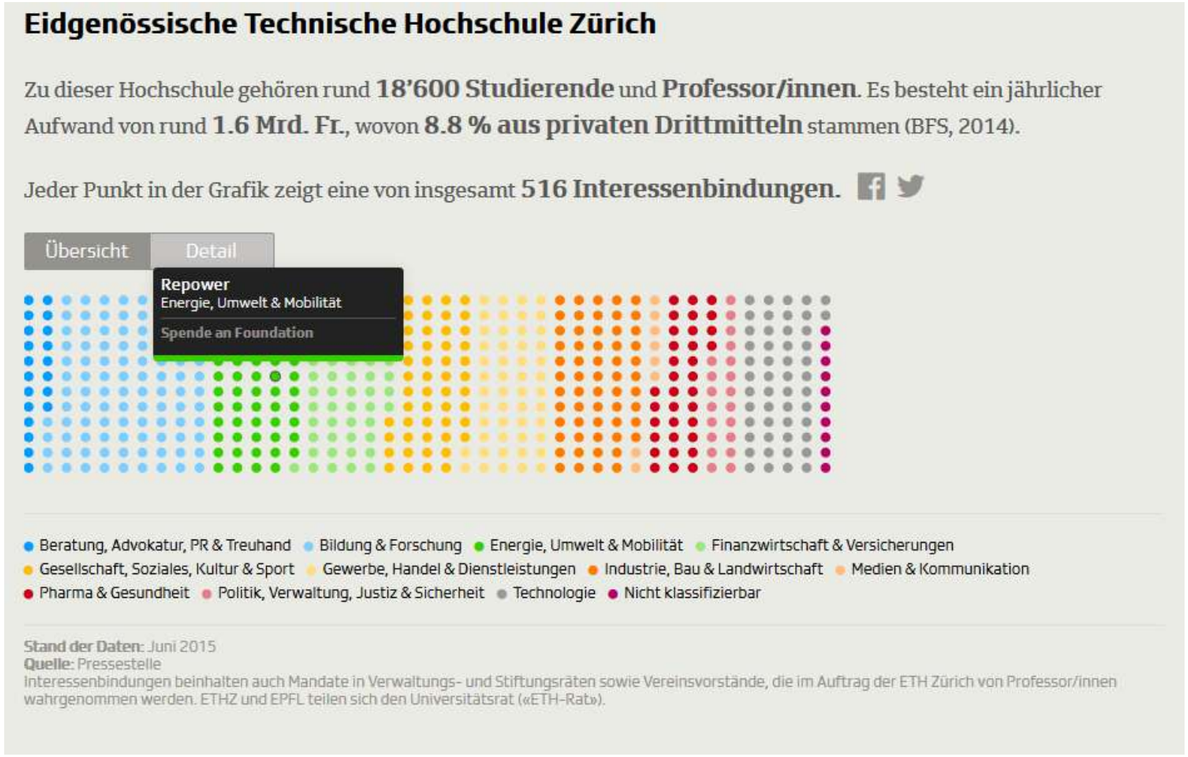
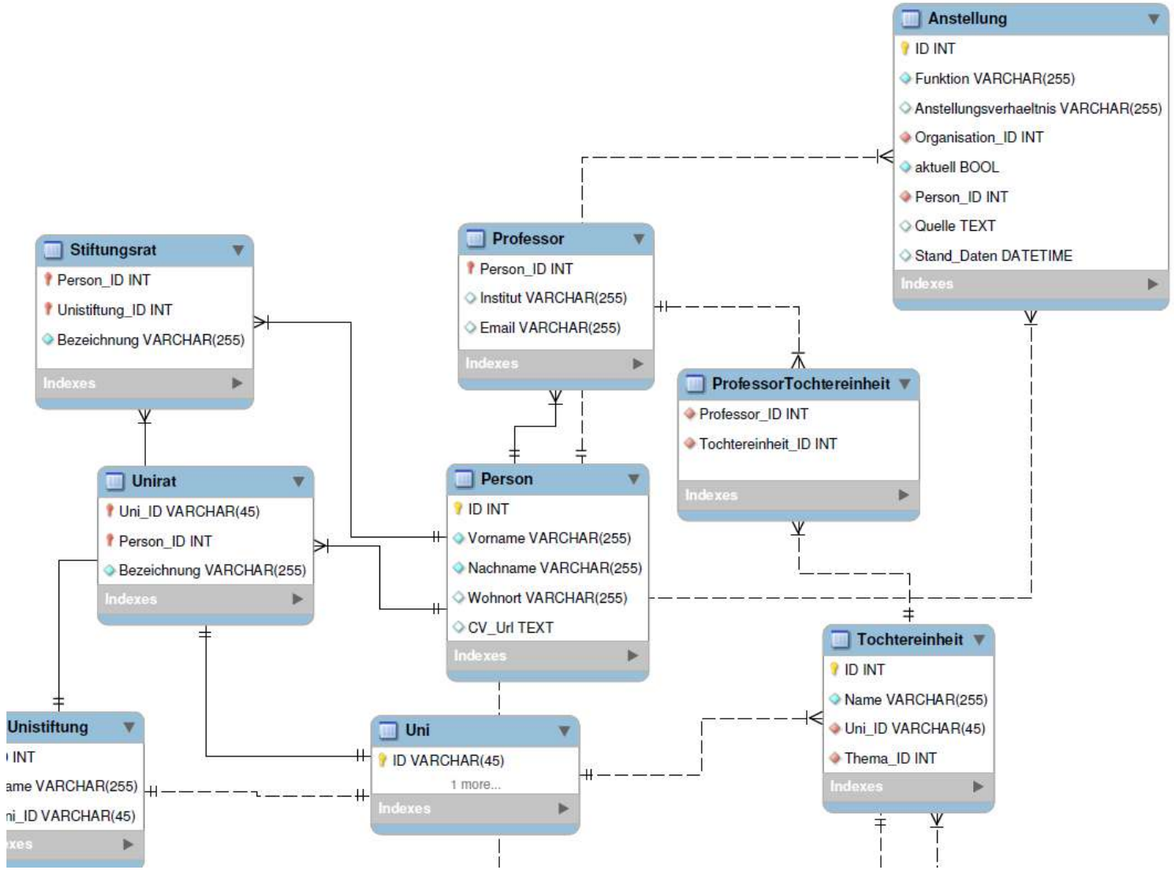
Your first database

INTRODUCTION TO RELATIONAL DATABASES IN SQL

Timo Grossenbacher
Data Journalist

Investigating universities in Switzerland





A relational database:

- real-life *entities* become *tables*
- reduced redundancy
- data integrity by *relationships*

- e.g. `professors`, `universities`, `companies`
- e.g. only one entry in `companies` for the bank "Credit Suisse"
- e.g. a `professor` can work at multiple `universities` and `companies`, a `company` can employ multiple `professors`

Throughout this course you will:

- work with the data I used for my investigation
- create a relational database from scratch
- learn three concepts:
  - *constraints*
  - *keys*
  - *referential integrity*

You'll need: Basic understanding of SQL, as taught in **Introduction to SQL**.

Your first duty: Have a look at the PostgreSQL database

```
SELECT table_schema, table_name
FROM information_schema.tables;
```

```
   table_schema    |              table_name
 ------------------+------------------------------
 pg_catalog        | pg_statistic
 pg_catalog        | pg_type
 pg_catalog        | pg_policy
 pg_catalog        | pg_authid
 pg_catalog        | pg_shadow
 public            | university_professors
 pg_catalog        | pg_settings
...
```

Have a look at the columns of a certain table

```
SELECT table_name, column_name, data_type
FROM information_schema.columns
WHERE table_name = 'pg_config';
```

```
 table_name | column_name | data_type
------------+-------------+-----------
 pg_config  | name        | text
 pg_config  | setting     | text
```

Let's do this.

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Tables: At the core of every database

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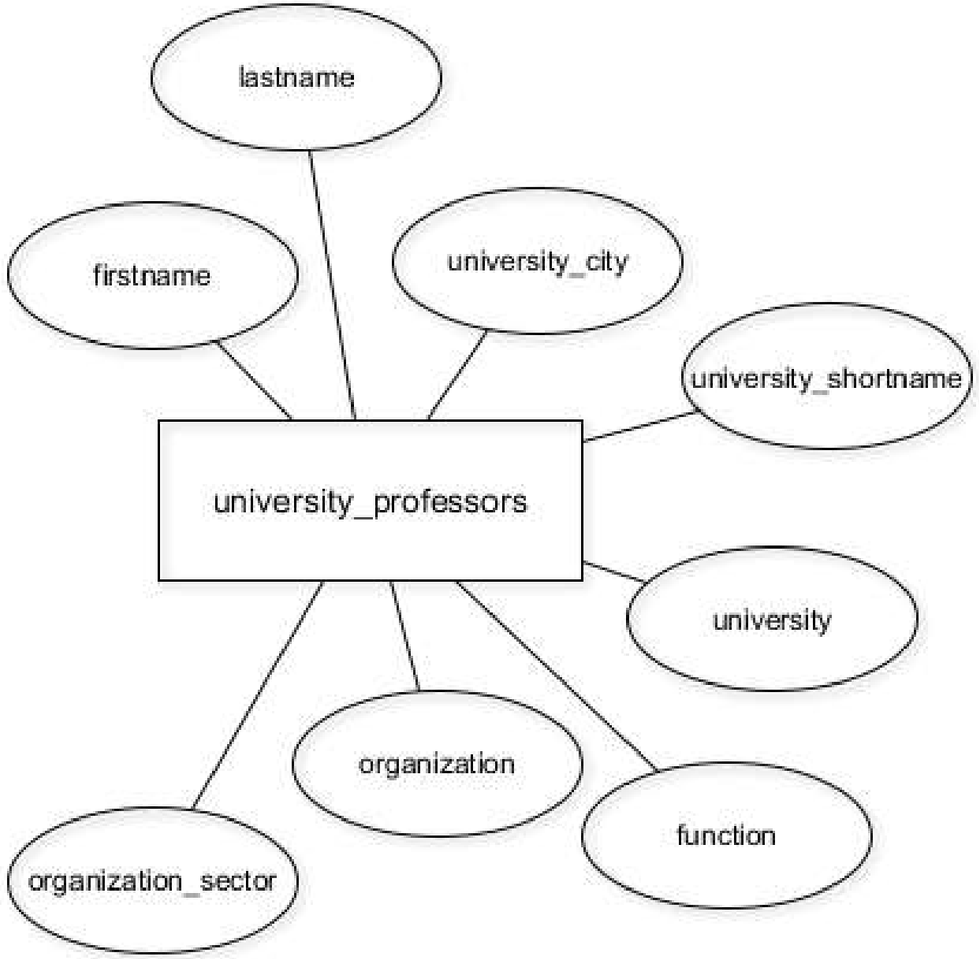
Redundancy in the university_professors table

```
SELECT *
FROM university_professors
LIMIT 3;
```

```
-[ RECORD 1 ]--------+--------------------------------------------------
firstname            | Karl
lastname             | Aberer
university           | ETH Lausanne
university_shortname | EPF
university_city      | Lausanne
function             | Chairman of L3S Advisory Board
organization         | L3S Advisory Board
organization_sector  | Education & research
-[ RECORD 2 ]--------+--------------------------------------------------
firstname            | Karl
lastname             | Aberer
university           | ETH Lausanne
university_shortname | EPF
university_city      | Lausanne
function             | Member Conseil of Zeno-Karl Schindler Foundation
organization         | Zeno-Karl Schindler Foundation
organization_sector  | Education & research
-[ RECORD 3 ]--------+--------------------------------------------------
firstname            | Karl
lastname             | Aberer
(truncated)
function             | Member of Conseil Fondation IDIAP
organization         | Fondation IDIAP
(truncated)
```

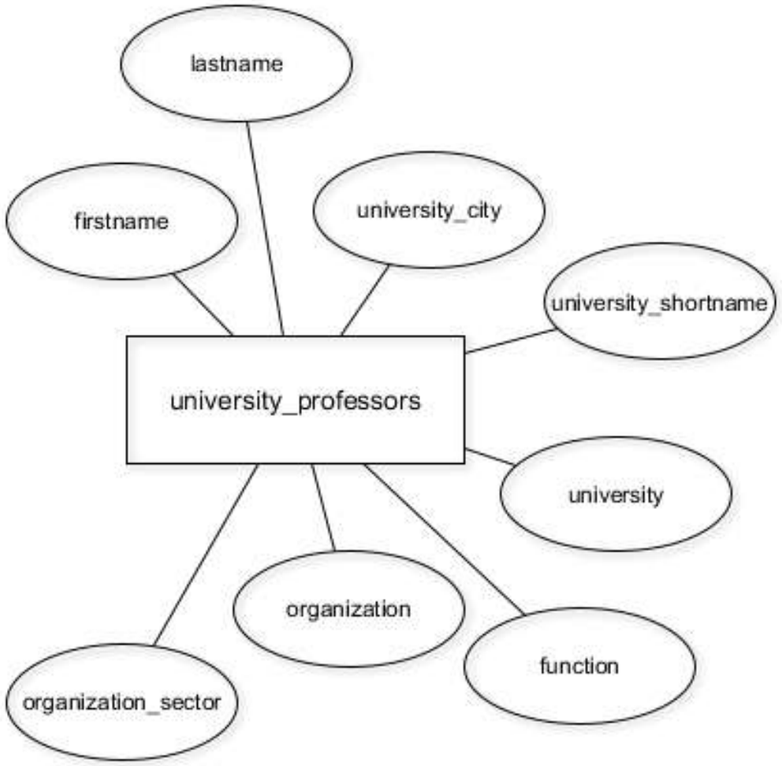
```
-[ RECORD 1 ]--------+-----------------------------------------
firstname            | Karl
lastname             | Aberer
university           | ETH Lausanne
university_shortname | EPF
university_city      | Lausanne
function             | Chairman of L3S Advisory Board
organisation         | L3S Advisory Board
organisation_sector  | Education & research
-[ RECORD 2 ]--------+-----------------------------------------
firstname            | Karl
lastname             | Aberer
university           | ETH Lausanne
university_shortname | EPF
university_city      | Lausanne
function             | Member Conseil of Zeno-Karl Schindler Foundation
organisation         | Zeno-Karl Schindler Foundation
organisation_sector  | Education & research
-[ RECORD 3 ]--------+-----------------------------------------
firstname            | Karl
lastname             | Aberer
(truncated)
function             | Member of Conseil Fondation IDIAP
organisation         | Fondation IDIAP
(truncated)
```

Currently: One "entity type" in the database

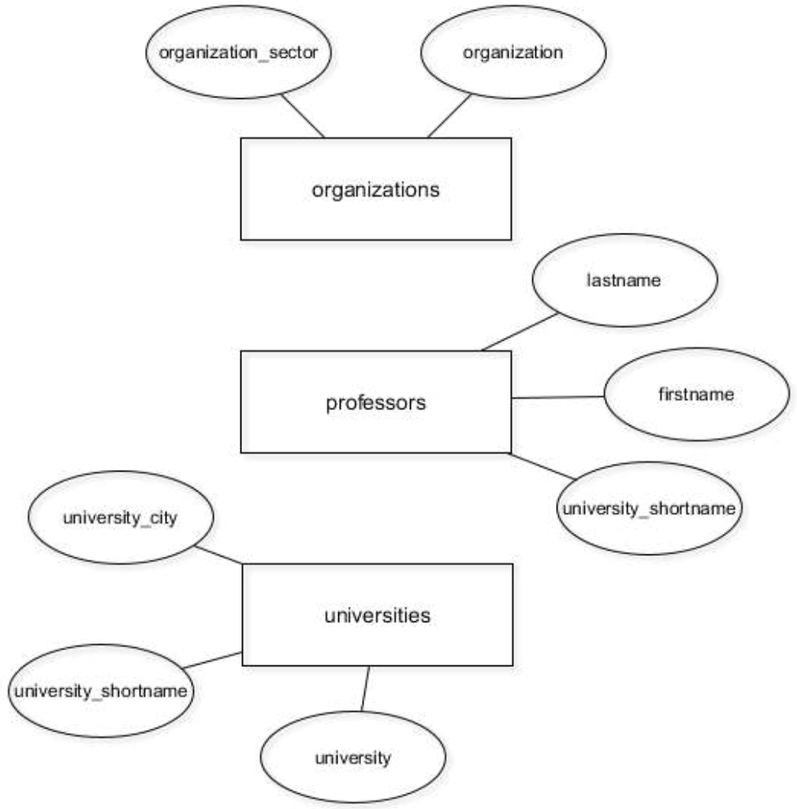


A better database model with three entity types

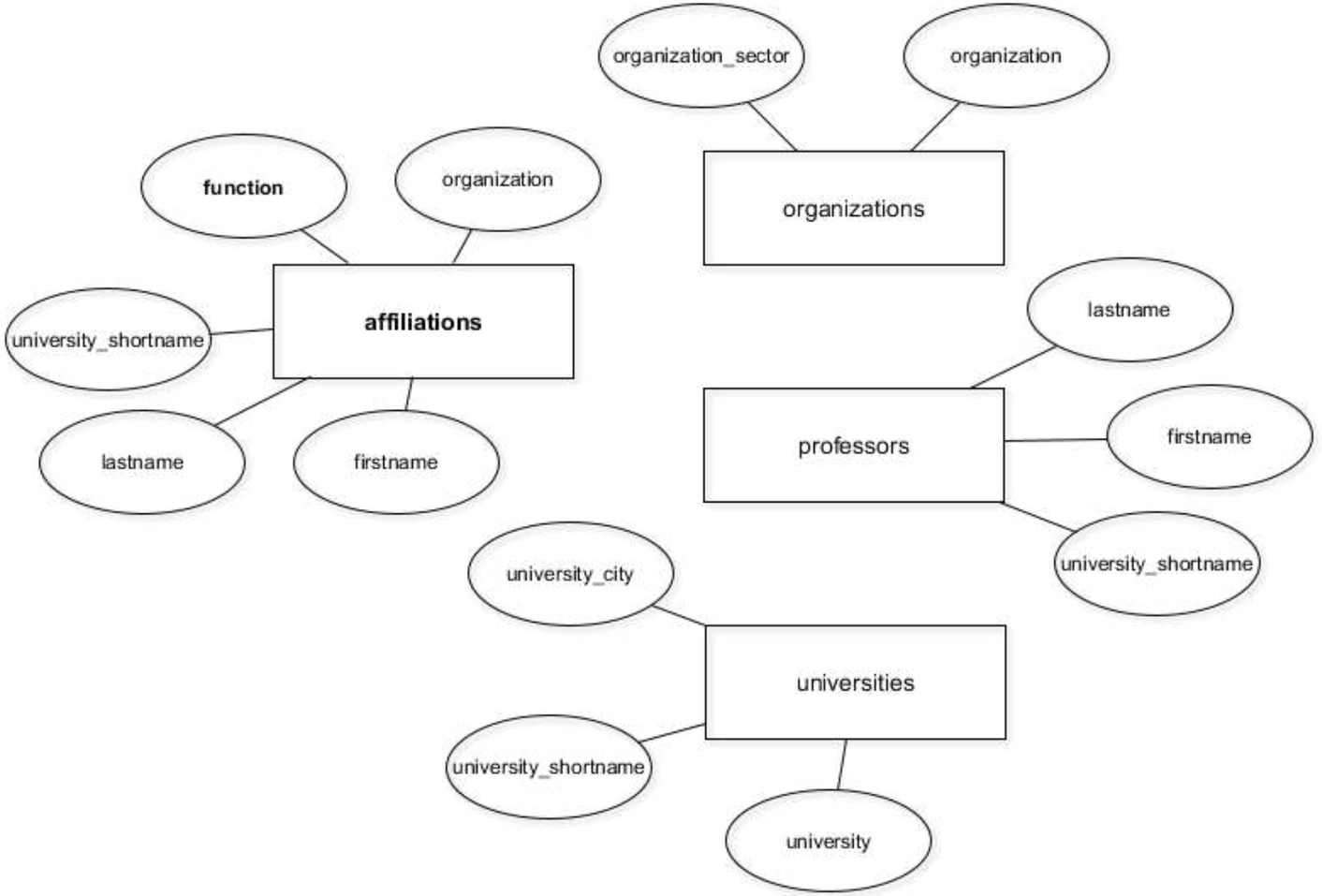
Old:



New:



A better database model with four entity types

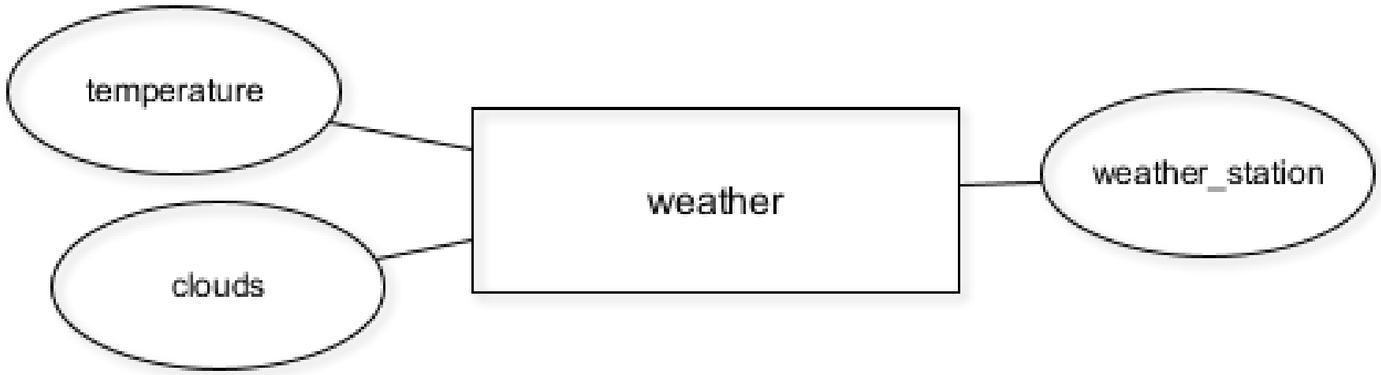


Create new tables with CREATE TABLE

```
CREATE TABLE table_name (
 column_a data_type,
 column_b data_type,
 column_c data_type
);
```

Create new tables with CREATE TABLE

```
CREATE TABLE weather (
 clouds text,
 temperature numeric,
 weather_station char(5)
);
```



Let's practice!

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Update your database as the structure changes

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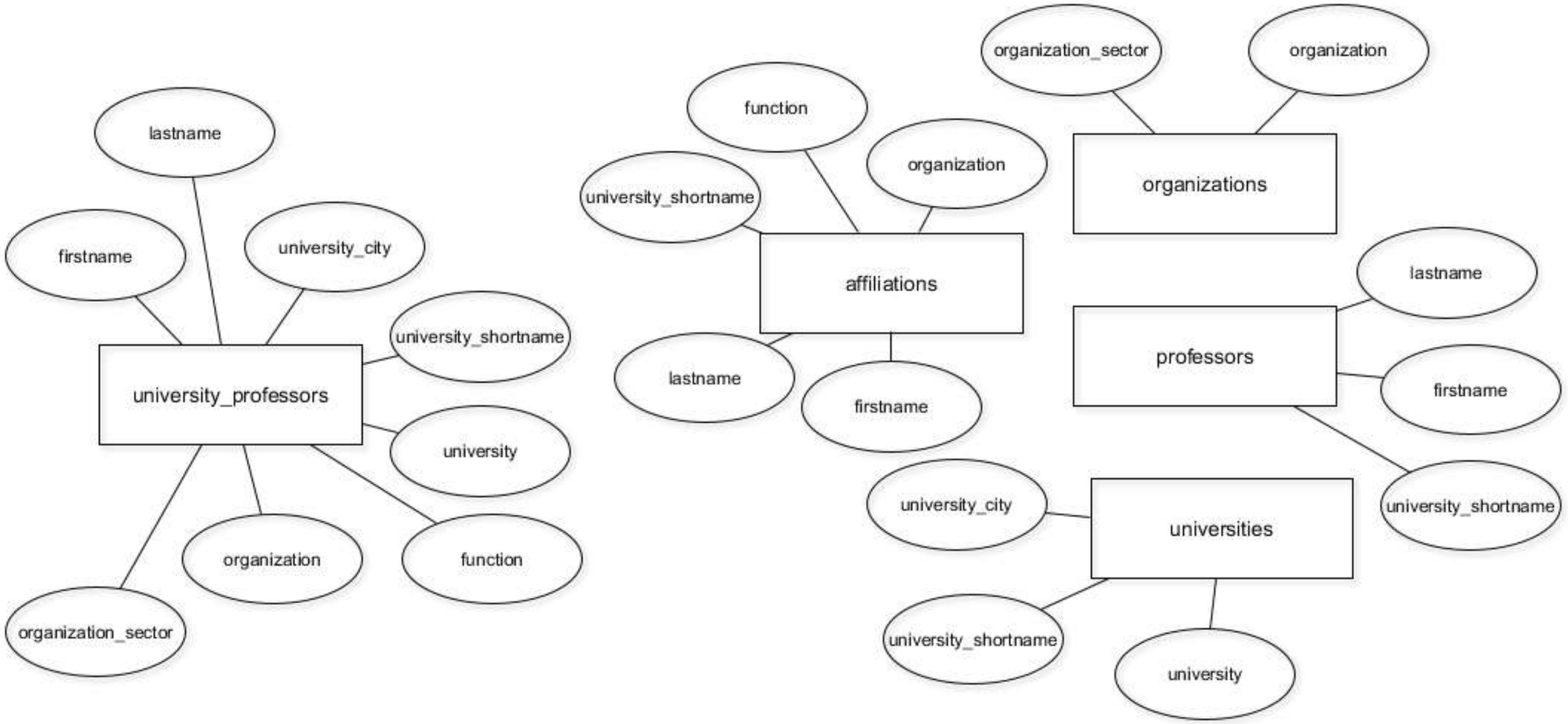
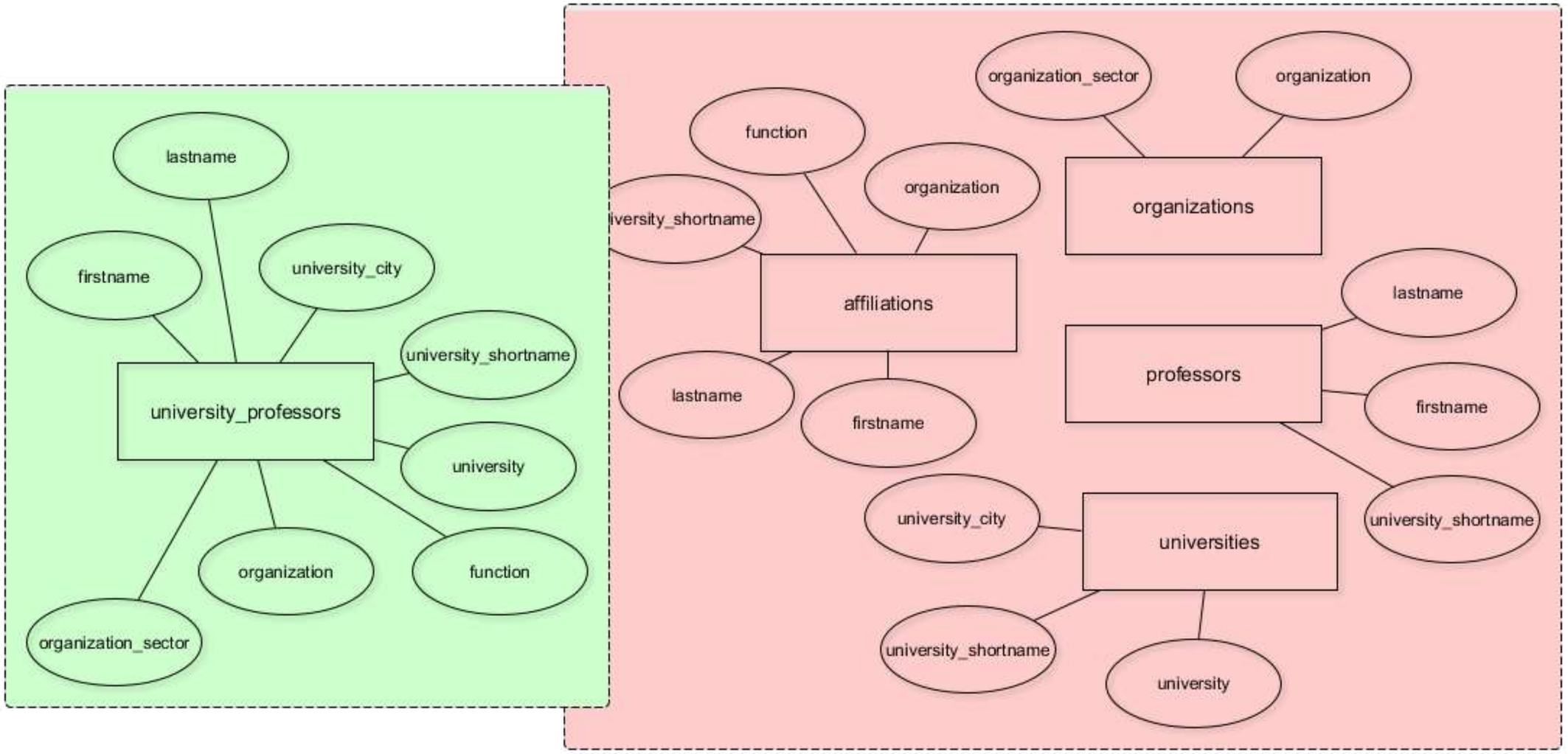
The current database model



The current database model



Only store DISTINCT data in the new tables

```
SELECT COUNT(*)
FROM university_professors;
```

```
count
-----
1377
```

```
SELECT COUNT(DISTINCT organization)
FROM university_professors;
```

```
count
-----
1287
```

INSERT DISTINCT records INTO the new tables

```
INSERT INTO organizations
SELECT DISTINCT organization,
    organization_sector
FROM university_professors;
```

```
Output: INSERT 0 1287
```

```
INSERT INTO organizations
SELECT organization,
    organization_sector
FROM university_professors;
```

```
Output: INSERT 0 1377
```

The INSERT INTO statement

```
INSERT INTO table_name (column_a, column_b)
VALUES ("value_a", "value_b");
```

RENAME a COLUMN in affiliations

```
CREATE TABLE affiliations (
 firstname text,
 lastname text,
 university_shortname text,
 function text,
 organisation text
);
```

```
ALTER TABLE table_name
RENAME COLUMN old_name TO new_name;
```

DROP a COLUMN in affiliations

```
CREATE TABLE affiliations (
 firstname text,
 lastname text,
 university_shortname text,
 function text,
 organization text
);
```

```
ALTER TABLE table_name
DROP COLUMN column_name;
```

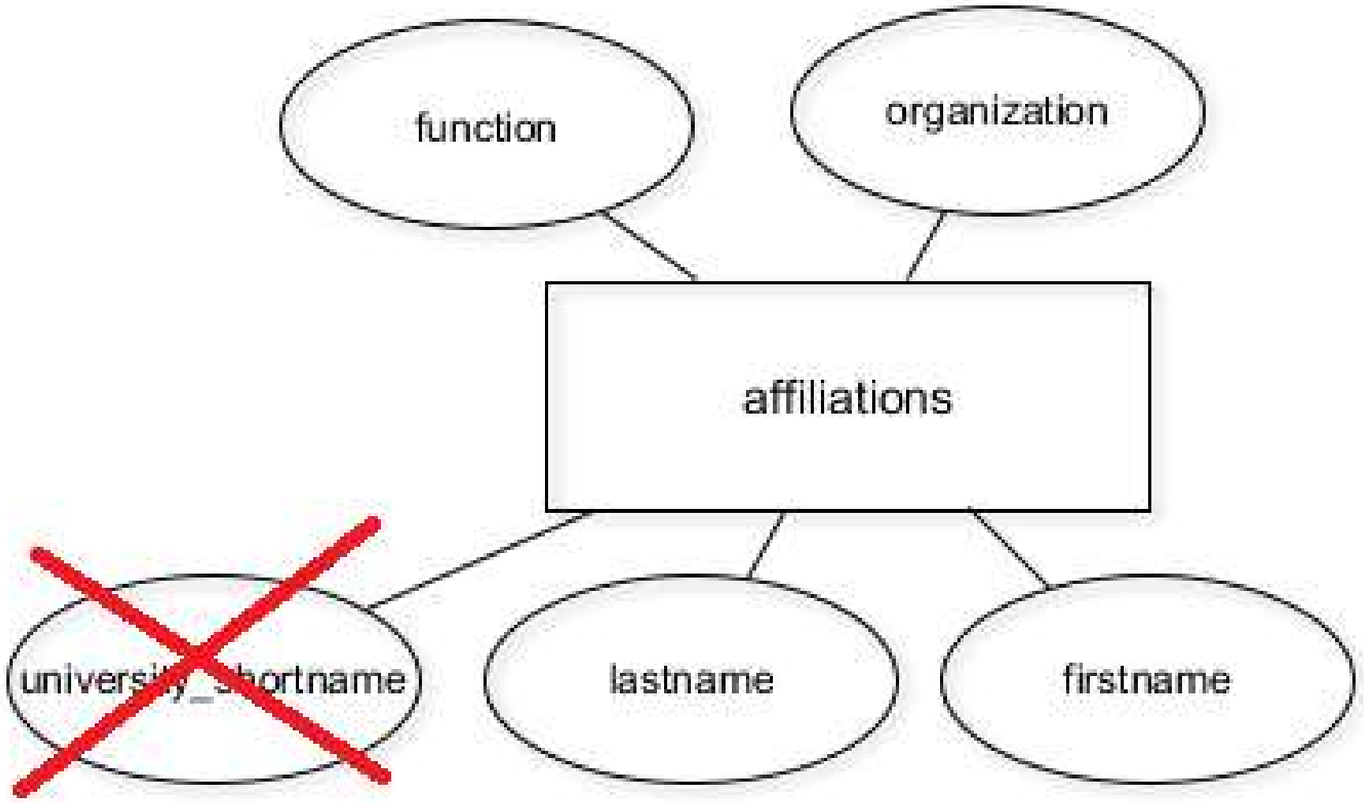
```
SELECT DISTINCT firstname, lastname,
    university_shortname
FROM university_professors
ORDER BY lastname;
```

```
-[ RECORD 1 ]--------+-------------
firstname            | Karl
lastname             | Aberer
university_shortname | EPF
-[ RECORD 2 ]--------+-------------
firstname            | Reza Shokrollah
lastname             | Abhari
university_shortname | ETH
-[ RECORD 3 ]--------+-------------
firstname            | Georges
lastname             | Abou Jaoudé
university_shortname | EPF
(truncated)

(551 records)
```

```
SELECT DISTINCT firstname, lastname
FROM university_professors
ORDER BY lastname;
```

```
-[ RECORD 1 ]----------------------
firstname | Karl
lastname  | Aberer
-[ RECORD 2 ]----------------------
firstname | Reza Shokrollah
lastname  | Abhari
-[ RECORD 3 ]----------------------
firstname | Georges
lastname  | Abou Jaoudé
(truncated)

(551 records)
```

A professor is uniquely identified by firstname, lastname only



Let's get to work!

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Better data quality with constraints

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Integrity constraints

1. **Attribute constraints**, e.g. data types on columns (Chapter 2)
2. **Key constraints**, e.g. primary keys (Chapter 3)
3. **Referential integrity constraints**, enforced through foreign keys (Chapter 4)

Why constraints?

- Constraints give the data structure
- Constraints help with consistency, and thus data quality
- Data quality is a business advantage / data science prerequisite
- Enforcing is difficult, but PostgreSQL helps

Data types as attribute constraints

Name	Aliases	Description
`bigint`	`int8`	signed eight-byte integer
`bigserial`	`serial8`	autoincrementing eight-byte integer
`bit [ (n) ]`		fixed-length bit string
`bit varying [ (n) ]`	`varbit [ (n) ]`	variable-length bit string
`boolean`	`bool`	logical Boolean (true/false)
`box`		rectangular box on a plane
`bytea`		binary data ("byte array")
`character [ (n) ]`	`char [ (n) ]`	fixed-length character string
`character varying [ (n) ]`	`varchar [ (n) ]`	variable-length character string
`cidr`		IPv4 or IPv6 network address

From the PostgreSQL documentation.

Dealing with data types (casting)

```
CREATE TABLE weather (
 temperature integer,
 wind_speed text);
SELECT temperature * wind_speed AS wind_chill
FROM weather;
```

```
operator does not exist: integer * text
HINT: No operator matches the given name and argument type(s).
You might need to add explicit type casts.
```

```
SELECT temperature * CAST(wind_speed AS integer) AS wind_chill
FROM weather;
```

Let's practice!

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Working with data types

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Working with data types

- Enforced on columns (i.e. attributes)
- Define the so-called "domain" of a column
- Define what operations are possible
- Enforce consistent storage of values

The most common types

- `text` : character strings of any length
- `varchar [ (x) ]` : a maximum of `x` characters
- `char [ (x) ]` : a fixed-length string of `x` characters
- `boolean` : can only take three states, e.g. `TRUE` , `FALSE` and `NULL` (unknown)

From the **PostgreSQL documentation**.

The most common types (cont'd.)

- `date`, `time` and `timestamp`: various formats for date and time calculations
- `numeric`: arbitrary precision numbers, e.g. `3.1457`
- `integer`: whole numbers in the range of `-2147483648` and `+2147483647`

From the **PostgreSQL documentation**.

Specifying types upon table creation

```
CREATE TABLE students (
 ssn integer,
 name varchar(64),
 dob date,
 average_grade numeric(3, 2), -- e.g. 5.54
 tuition_paid boolean
);
```

Alter types after table creation

```
ALTER TABLE students
ALTER COLUMN name
TYPE varchar(128);
```

```
ALTER TABLE students
ALTER COLUMN average_grade
TYPE integer
-- Turns 5.54 into 6, not 5, before type conversion
USING ROUND(average_grade);
```

Let's apply this!

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The not-null and unique constraints

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The not-null constraint

- Disallow `NULL` values in a certain column
- Must hold true for the current state
- Must hold true for any future state

What does NULL mean?

- unknown
- does not exist
- does not apply
- ...

What does NULL mean? An example

```
CREATE TABLE students (
 ssn integer not null,
 lastname varchar(64) not null,
 home_phone integer,
 office_phone integer
);
```

```
NULL != NULL
```

How to add or remove a not-null constraint

When creating a table...

```
CREATE TABLE students (
 ssn integer not null,
 lastname varchar(64) not null,
 home_phone integer,
 office_phone integer
);
```

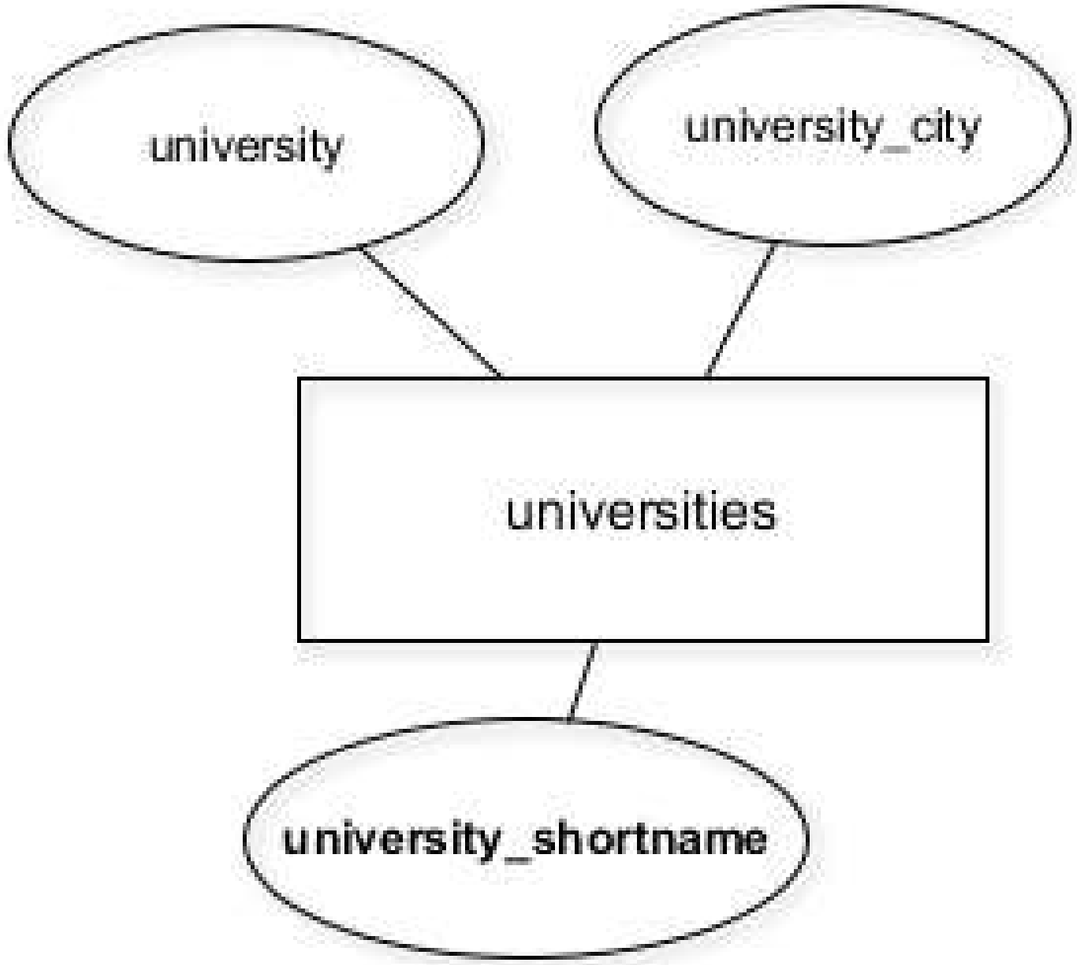
After the table has been created...

```
ALTER TABLE students
ALTER COLUMN home_phone
SET NOT NULL;
```

```
ALTER TABLE students
ALTER COLUMN ssn
DROP NOT NULL;
```

The unique constraint

- Disallow duplicate values in a column
- Must hold true for the current state
- Must hold true for any future state



Adding unique constraints

```
CREATE TABLE table_name (
 column_name UNIQUE
);
```

```
ALTER TABLE table_name
ADD CONSTRAINT some_name UNIQUE(column_name);
```

Let's apply this to the database!

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Keys and superkeys

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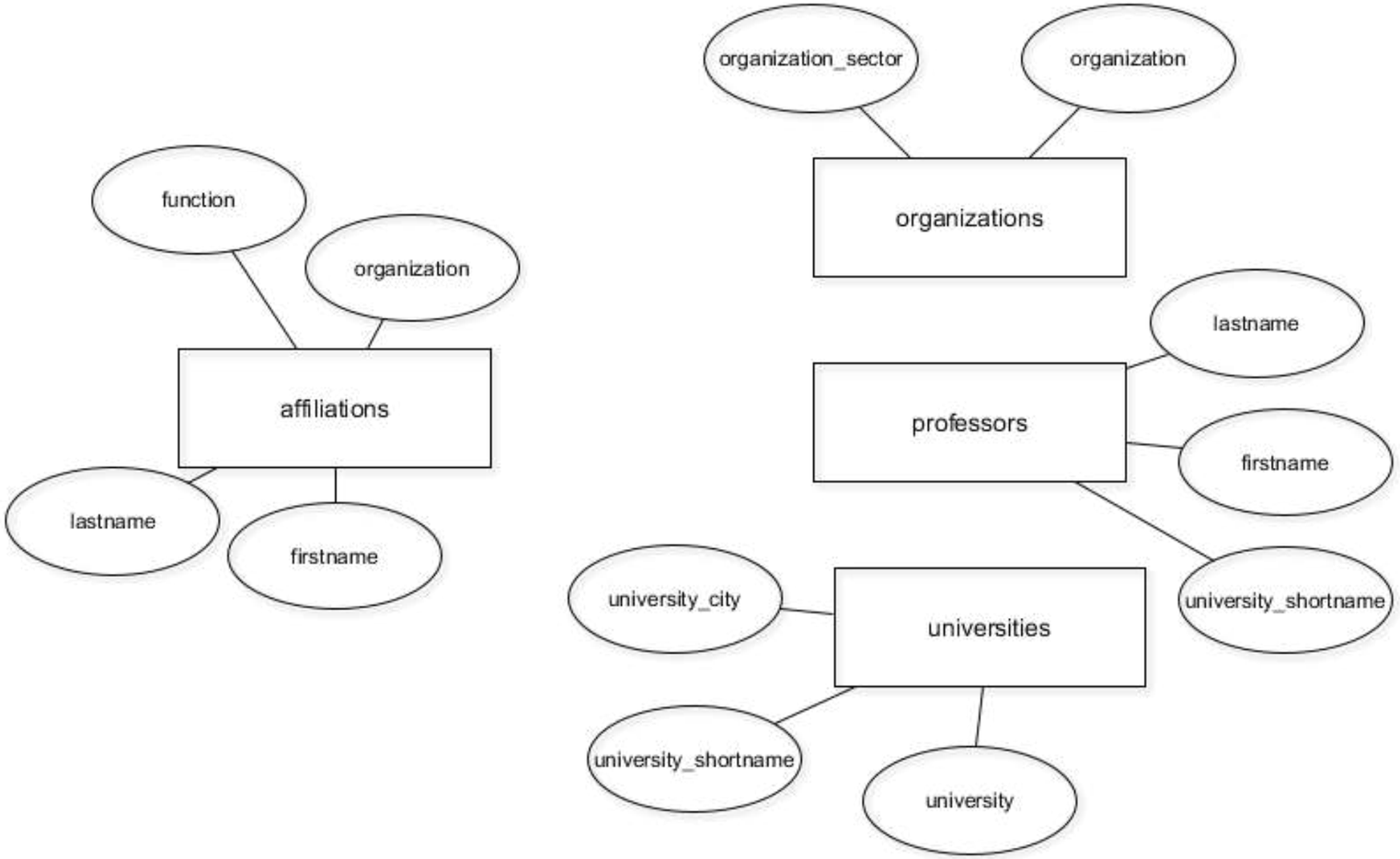
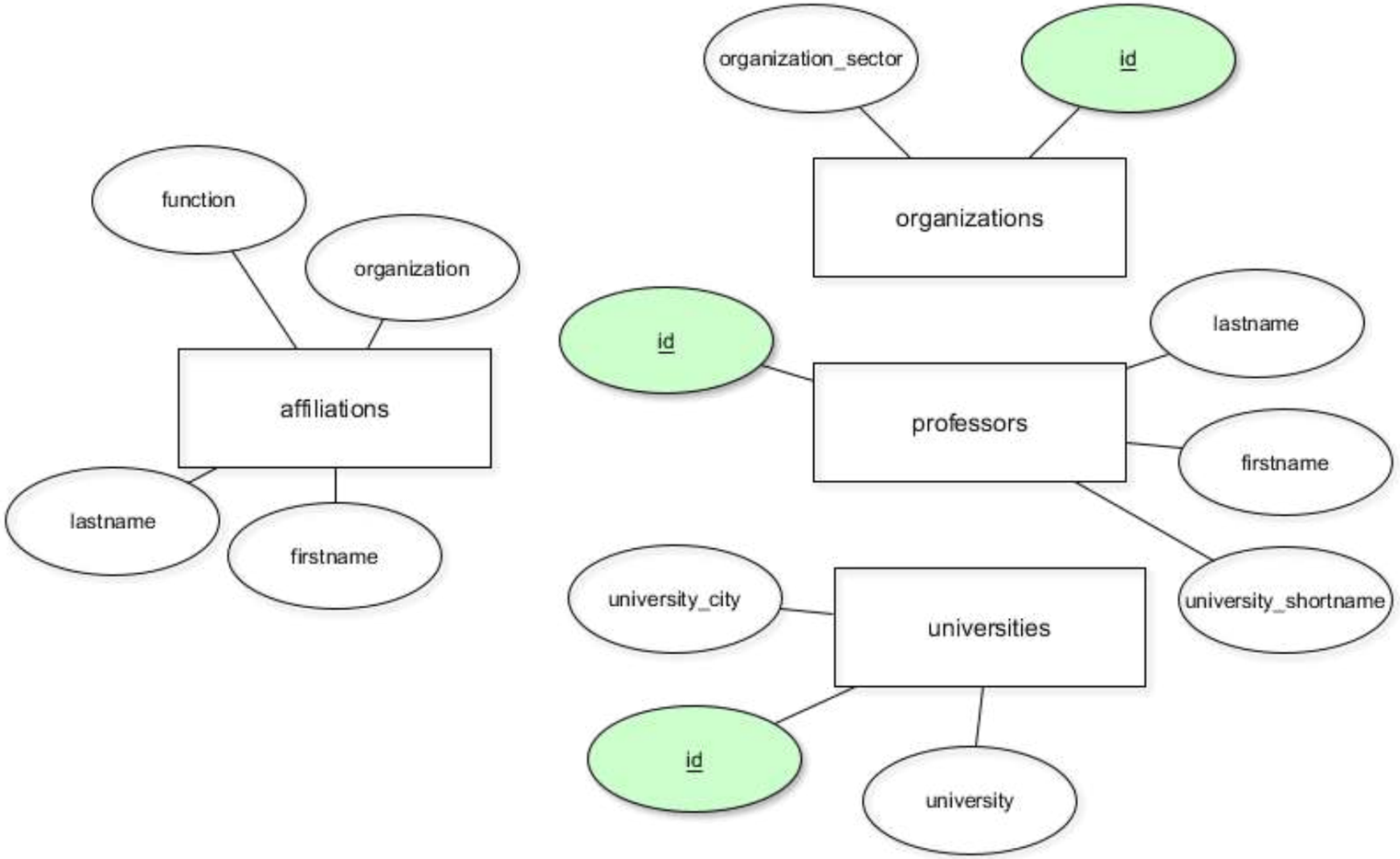
The current database model



The database model with primary keys



What is a key?

- Attribute(s) that identify a record uniquely
- As long as attributes can be removed: **superkey**
- If no more attributes can be removed: minimal superkey or **key**

license_no	serial_no	make	model	year
Texas ABC-739	A69352	Ford	Mustang	2
Florida TVP-347	B43696	Oldsmobile	Cutlass	5
New York MPO-22	X83554	Oldsmobile	Delta	1
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	4
Texas RSK-629	U028365	Jaguar	XJS	4

SK1 = {license_no, serial_no, make, model, year}

SK2 = {license_no, serial_no, make, model}

SK3 = {make, model, year}, SK4 = {license_no, serial_no}, SKi, ..., SKn

Adapted from Elmasri, Navathe (2011): Fundamentals of Database Systems, 6th Ed., Pearson

license_no	serial_no	make	model	year
Texas ABC-739	A69352	Ford	Mustang	2
Florida TVP-347	B43696	Oldsmobile	Cutlass	5
New York MPO-22	X83554	Oldsmobile	Delta	1
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	4
Texas RSK-629	U028365	Jaguar	XJS	4

K1 = {license_no}; K2 = {serial_no}; K3 = {model}; K4 = {make, year}

- K1 to 3 only consist of one attribute
- Removing either "make" or "year" from K4 would result in duplicates
- Only one candidate key can be the *chosen* key

Let's discover some keys!

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Primary keys

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Primary keys

- One primary key per database table, chosen from candidate keys
- Uniquely identifies records, e.g. for referencing in other tables
- Unique and not-null constraints both apply
- Primary keys are time-invariant: choose columns wisely!

Specifying primary keys

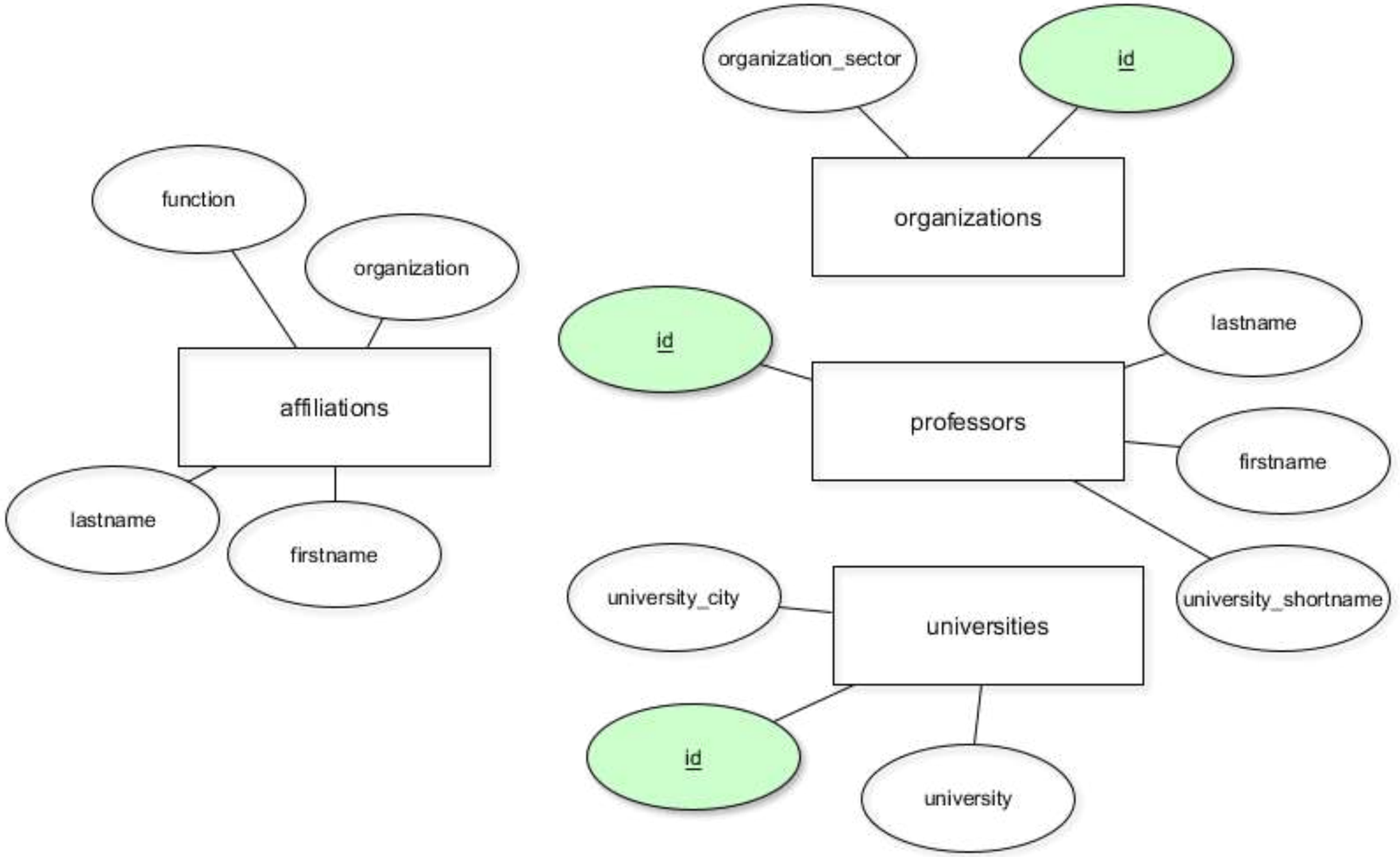
```
CREATE TABLE products (
    product_no integer UNIQUE NOT NULL,
    name text,
    price numeric
);
CREATE TABLE products (
    product_no integer PRIMARY KEY,
    name text,
    price numeric
);
```

```
CREATE TABLE example (
    a integer,
    b integer,
    c integer,
    PRIMARY KEY (a, c)
);
```

Taken from the PostgreSQL documentation.

Specifying primary keys (contd.)

```
ALTER TABLE table_name
ADD CONSTRAINT some_name PRIMARY KEY (column_name)
```


function
organization
affiliations
lastname
firstname
organization_sector
id
organizations
id
lastname
professors
firstname
university_city
universities
university_shortname
id
university

Let's practice!

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Surrogate keys

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Surrogate keys

- Primary keys should be built from as few columns as possible
- Primary keys should never change over time

license_no	serial_no	make	model	color
Texas ABC-739	A69352	Ford	Mustang	blue
Florida TVP-347	B43696	Oldsmobile	Cutlass	black
New York MPO-22	X83554	Oldsmobile	Delta	silver
California 432-TFY	C43742	Mercedes	190-D	champagne
California RSK-629	Y82935	Toyota	Camry	red
Texas RSK-629	U028365	Jaguar	XJS	blue

make	model	color
Ford	Mustang	blue
Oldsmobile	Cutlass	black
Oldsmobile	Delta	silver
Mercedes	190-D	champagne
Toyota	Camry	red
Jaguar	XJS	blue

Adding a surrogate key with serial data type

```
ALTER TABLE cars
ADD COLUMN id serial PRIMARY KEY;
INSERT INTO cars
VALUES ('Volkswagen', 'Blitz', 'black');
```

```
   make     |  model  | color       | id
 -----------+---------+-------------+-------------
 Ford       | Mustang | blue        | 1
 Oldsmobile | Cutlass | black       | 2
 Oldsmobile | Delta   | silver      | 3
 Mercedes   | 190-D   | champagne   | 4
 Toyota     | Camry   | red         | 5
 Jaguar     | XJS     | blue        | 6
 Volkswagen | Blitz   | black       | 7
```

Adding a surrogate key with serial data type (contd.)

```
INSERT INTO cars
VALUES ('Opel', 'Astra', 'green', 1);
```

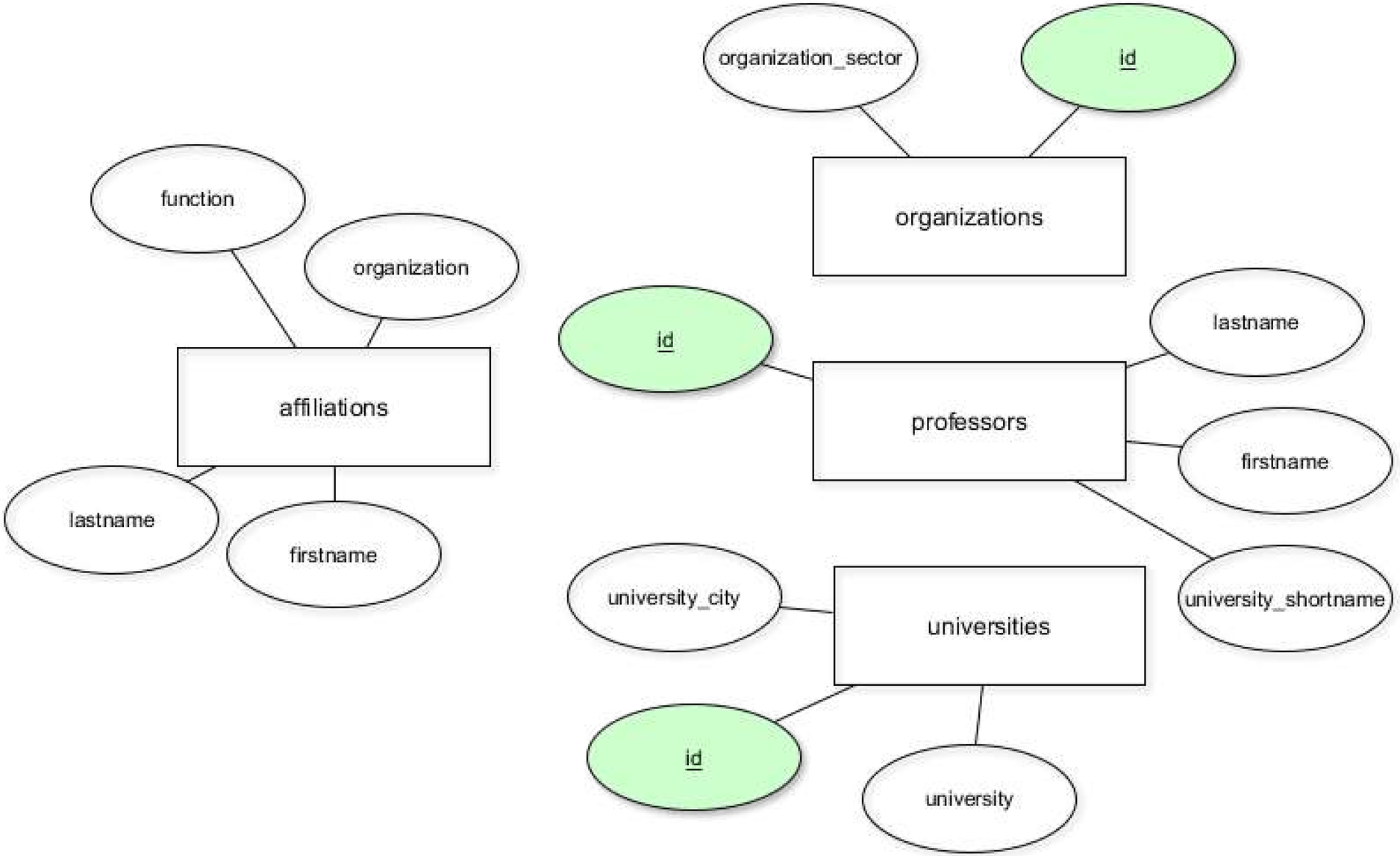
```
duplicate key value violates unique constraint "id_pkey"
DETAIL:  Key (id)=(1) already exists.
```

- "id" uniquely identifies records in the table – useful for referencing!

Another type of surrogate key

```
ALTER TABLE table_name
ADD COLUMN column_c varchar(256);

UPDATE table_name
SET column_c = CONCAT(column_a, column_b);
ALTER TABLE table_name
ADD CONSTRAINT pk PRIMARY KEY (column_c);
```


organization_sector
id
organizations
function
organization
affiliations
lastname
firstname
id
lastname
professors
firstname
university_city
universities
university_shortname
id
university

Let's try this!

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Model 1:N relationships with foreign keys

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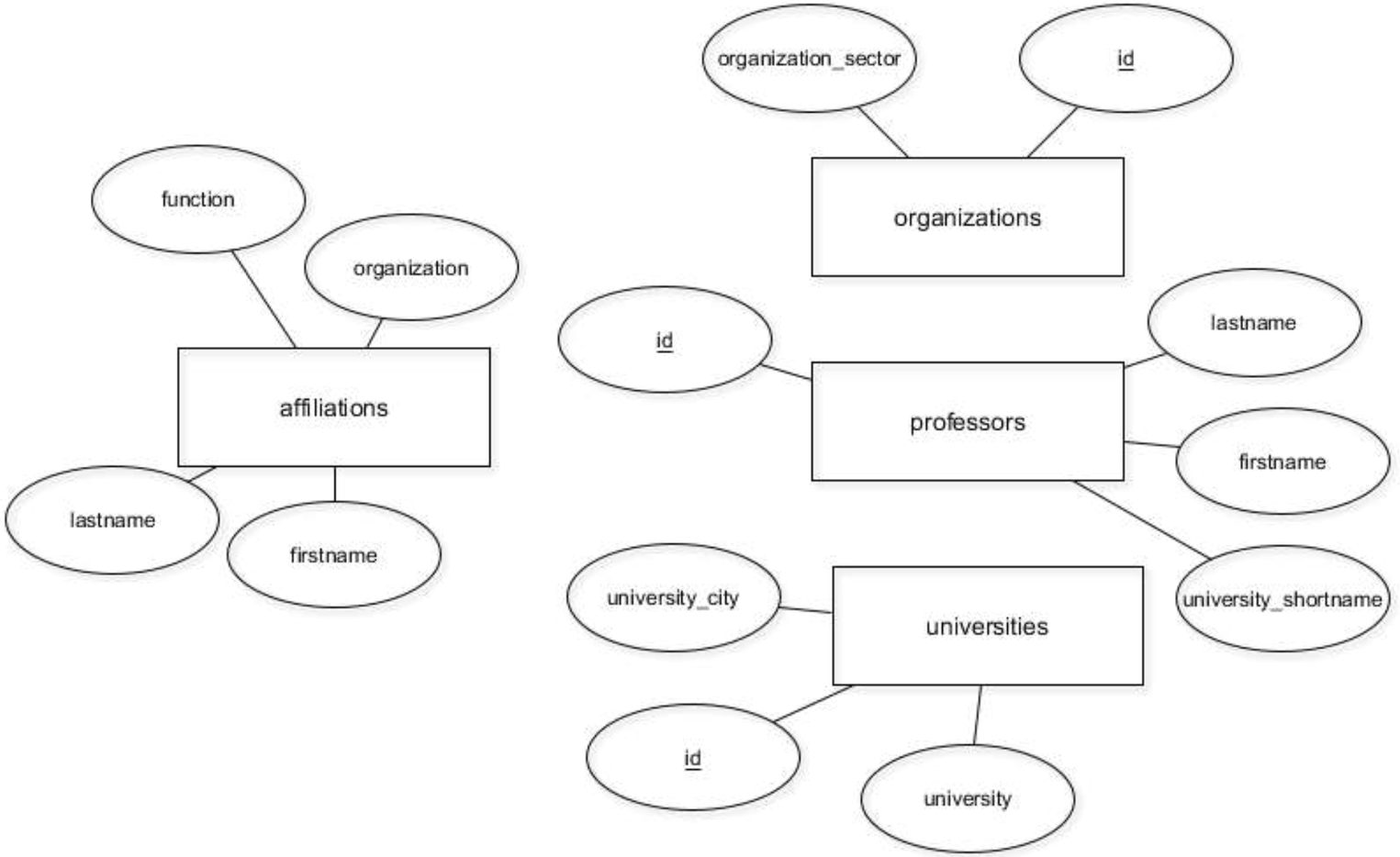
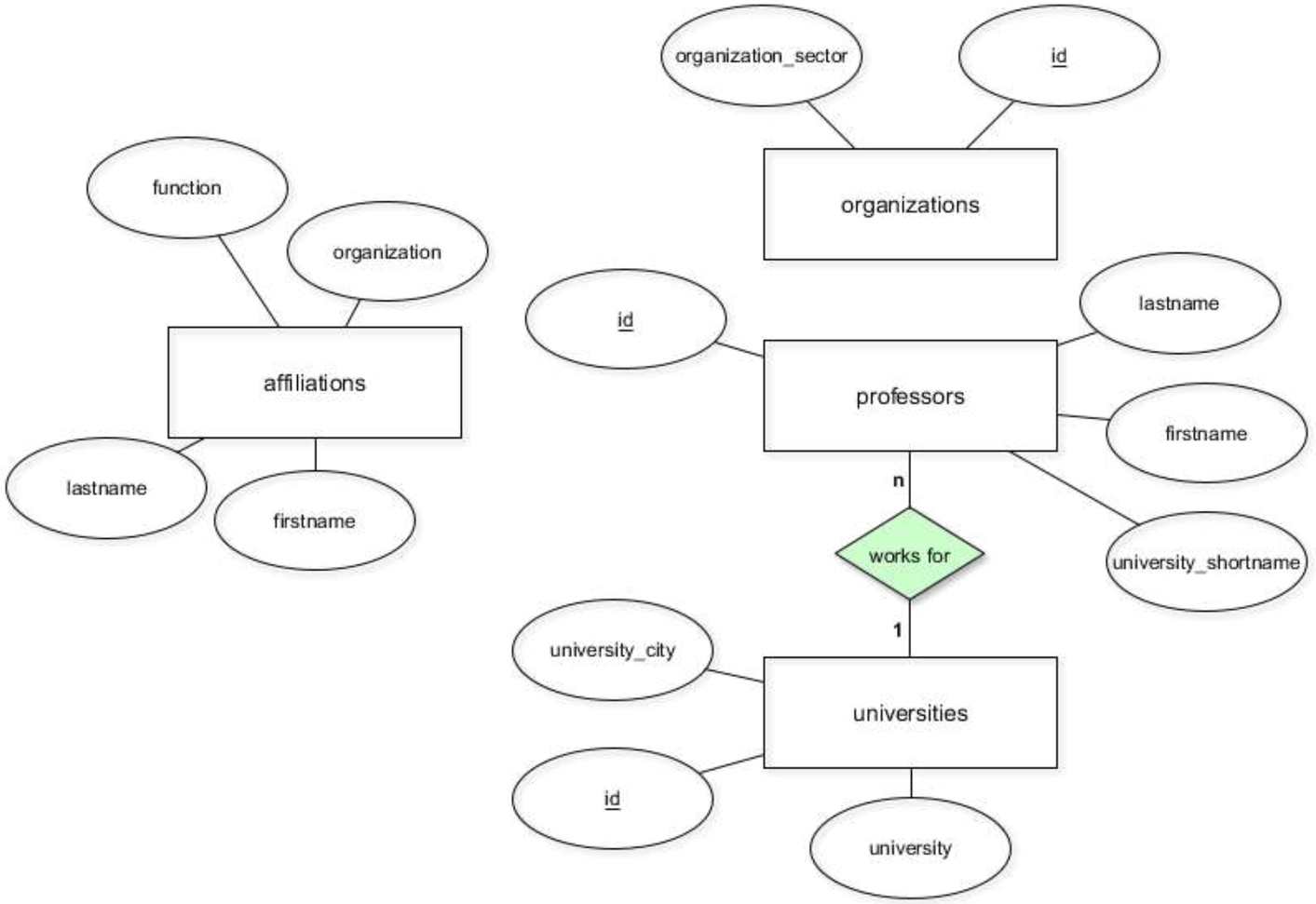
The current database model



The next database model



Implementing relationships with foreign keys

- A foreign key (FK) points to the primary key (PK) of another table
- Domain of FK must be equal to domain of PK
- Each value of FK must exist in PK of the other table (FK constraint or "referential integrity")
- FKs are not actual *keys*

A query

```
SELECT * FROM professors LIMIT 8;
```

```
id |    firstname    |  lastname   | university_s..
 --+-----------------+-------------+------------
 1 | Karl            | Aberer      | EPF
 2 | Reza Shokrollah | Abhari      | ETH
 3 | Georges         | Abou Jaoudé | EPF
 4 | Hugues          | Abriel      | UBE
 5 | Daniel          | Aebersold   | UBE
 6 | Marcelo         | Aebi        | ULA
 7 | Christoph       | Aebi        | UBE
 8 | Patrick         | Aebischer   | EPF
```

```
SELECT * FROM universities;
```

```
 id  |   university   | university_city
<hr />--+----------------+----------------
 EPF | ETH Lausanne   | Lausanne
 ETH | ETH Zürich     | Zurich
 UBA | Uni Basel      | Basel
 UBE | Uni Bern       | Bern
 UFR | Uni Freiburg   | Fribourg
 UGE | Uni Genf       | Geneva
 ULA | Uni Lausanne   | Lausanne
 UNE | Uni Neuenburg  | Neuchâtel
 USG | Uni St. Gallen | Saint Gallen
 USI | USI Lugano     | Lugano
 UZH | Uni Zürich     | Zurich
```

Specifying foreign keys

```
CREATE TABLE manufacturers (
 name varchar(255) PRIMARY KEY);

INSERT INTO manufacturers
VALUES ('Ford'), ('VW'), ('GM');
CREATE TABLE cars (
 model varchar(255) PRIMARY KEY,
 manufacturer_name varchar(255) REFERENCES manufacturers (name));

INSERT INTO cars
VALUES ('Ranger', 'Ford'), ('Beetle', 'VW');
```

```
-- Throws an error!
INSERT INTO cars
VALUES ('Tundra', 'Toyota');
```

Specifying foreign keys to existing tables

```
ALTER TABLE a
ADD CONSTRAINT a_fkey FOREIGN KEY (b_id) REFERENCES b (id);
```

Let's implement this!

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Model more complex relationships

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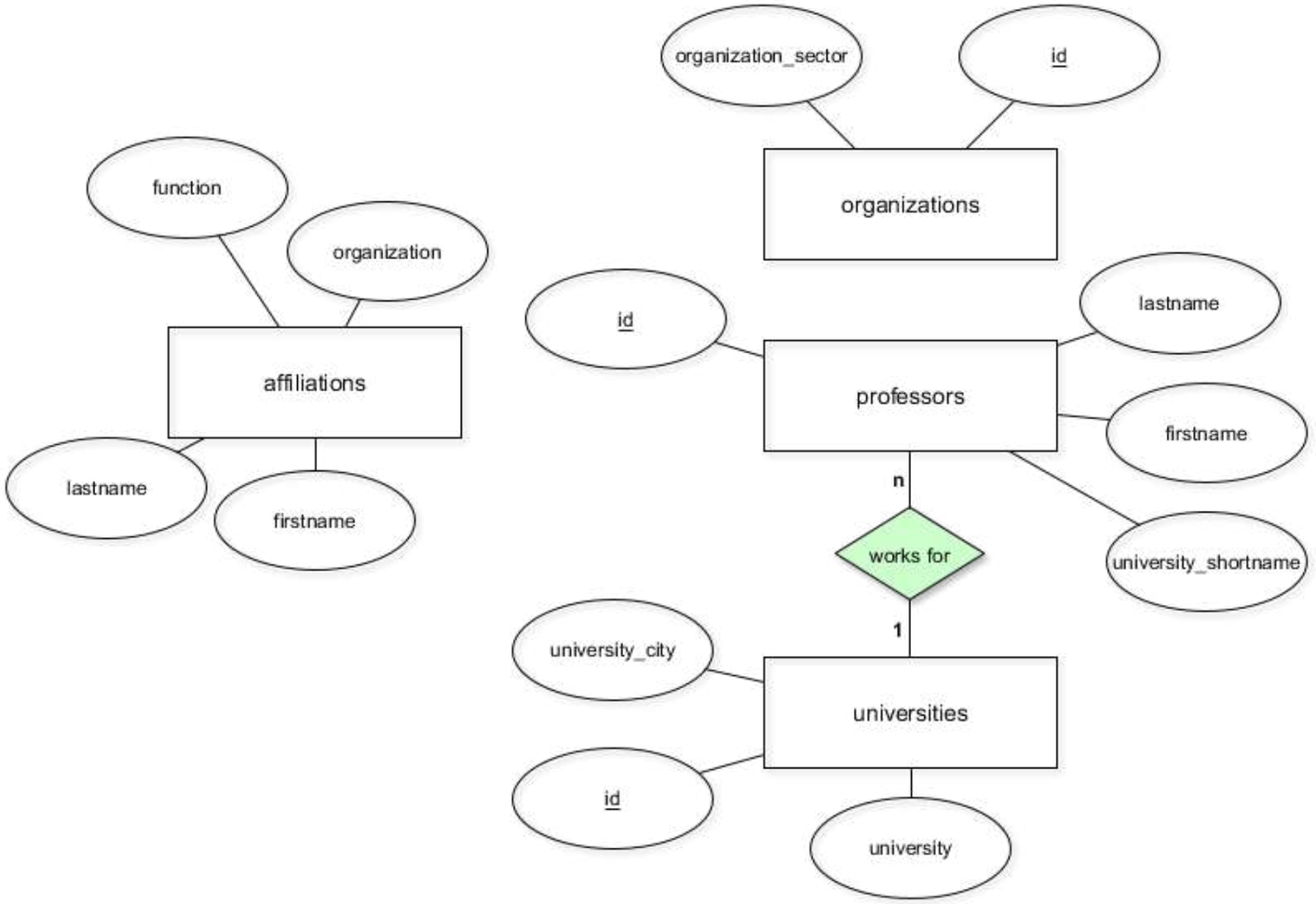
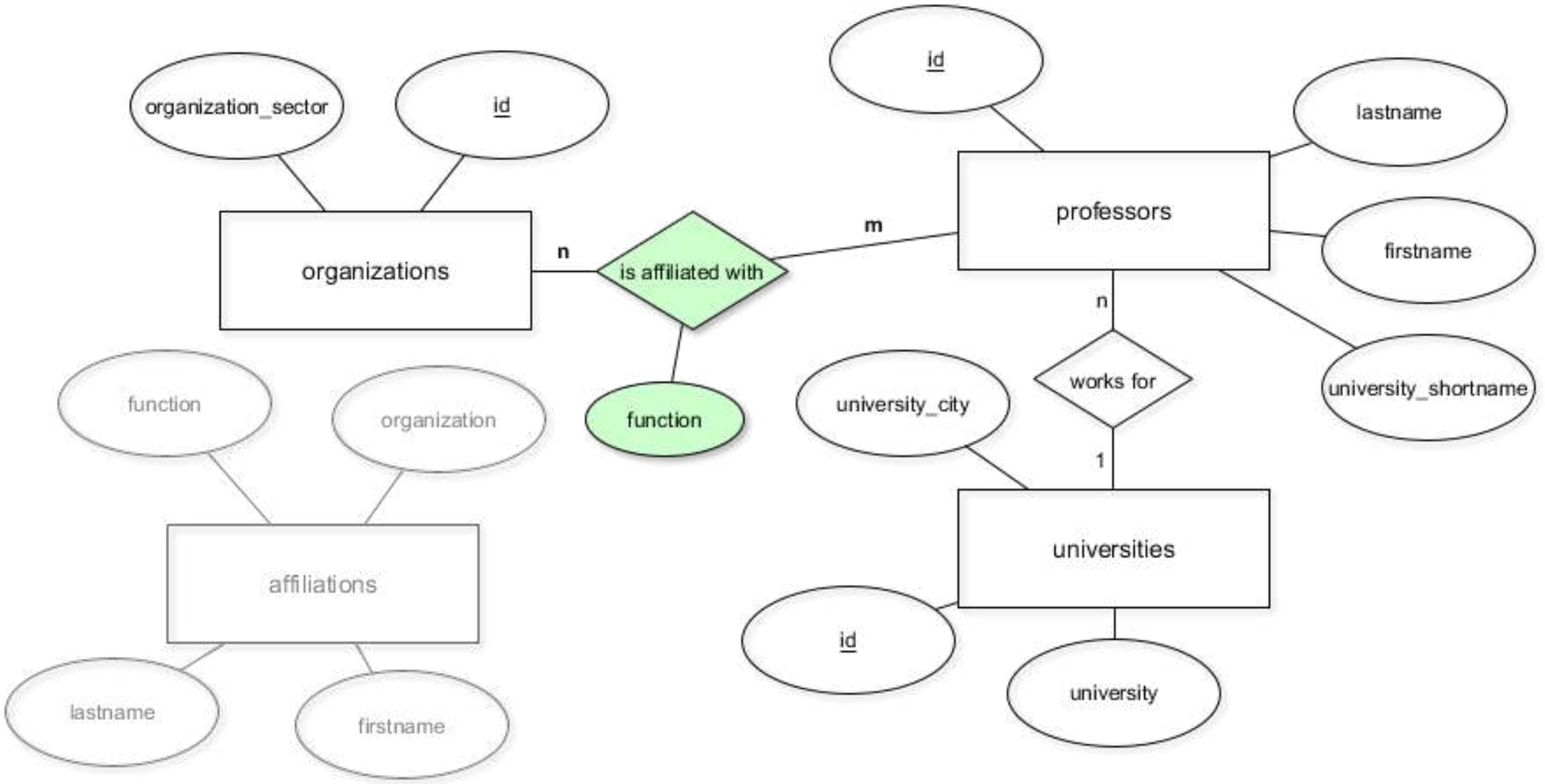
Timo Grossenbacher
Data Journalist

The current database model



The final database model



How to implement N:M-relationships

- Create a table
- Add foreign keys for every connected table
- Add additional attributes

```
CREATE TABLE affiliations (
 professor_id integer REFERENCES professors (id),
 organization_id varchar(256) REFERENCES organizations (id),
 function varchar(256)
);
```

- No primary key!
- Possible PK = {professor_id, organization_id, function}

Time to implement this!

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Referential integrity

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Data Journalist

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Referential integrity

- *A record referencing another table must refer to an existing record in that table*
- Specified between two tables
- Enforced through foreign keys

Referential integrity violations

Referential integrity from table A to table B is violated...

- ...if a record in table B that is referenced from a record in table A is deleted.
- ...if a record in table A referencing a non-existing record from table B is inserted.
- Foreign keys prevent violations!

Dealing with violations

```
CREATE TABLE a (
 id integer PRIMARY KEY,
 column_a varchar(64),
 ...,
 b_id integer REFERENCES b (id) ON DELETE NO ACTION
);
```

```
CREATE TABLE a (
 id integer PRIMARY KEY,
 column_a varchar(64),
 ...,
 b_id integer REFERENCES b (id) ON DELETE CASCADE
);
```

Dealing with violations, contd.

ON DELETE...

- ...NO ACTION: Throw an error
- ...CASCADE: Delete all referencing records
- ...RESTRICT: Throw an error
- ...SET NULL: Set the referencing column to NULL
- ...SET DEFAULT: Set the referencing column to its default value

Let's look at some examples!

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Roundup

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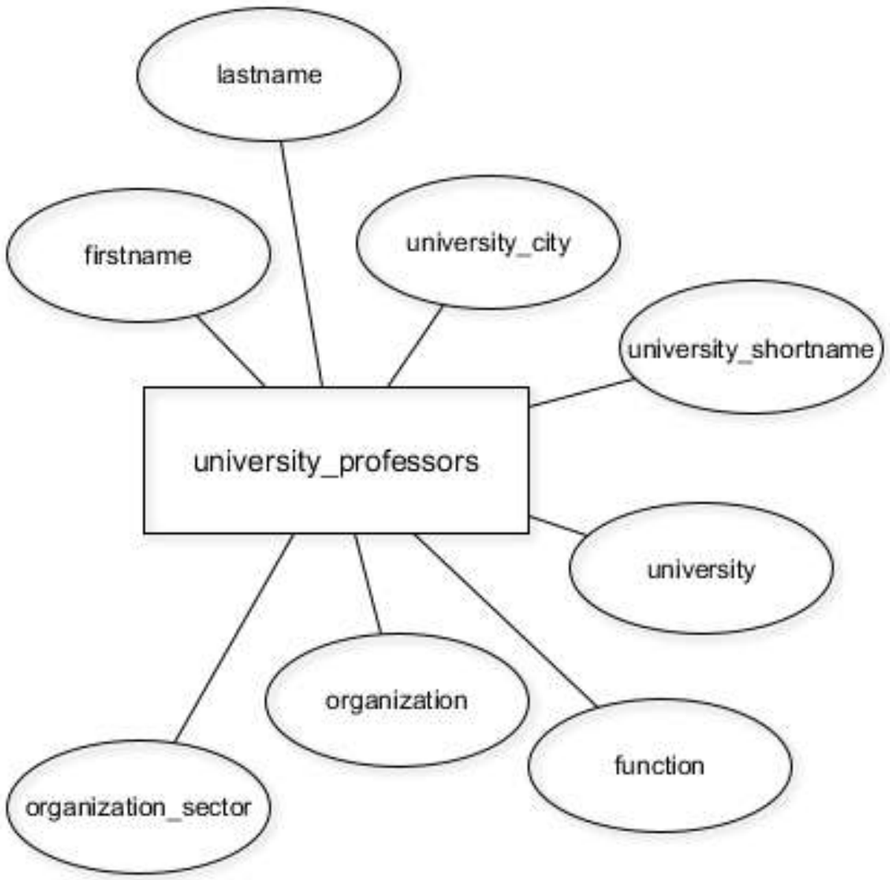
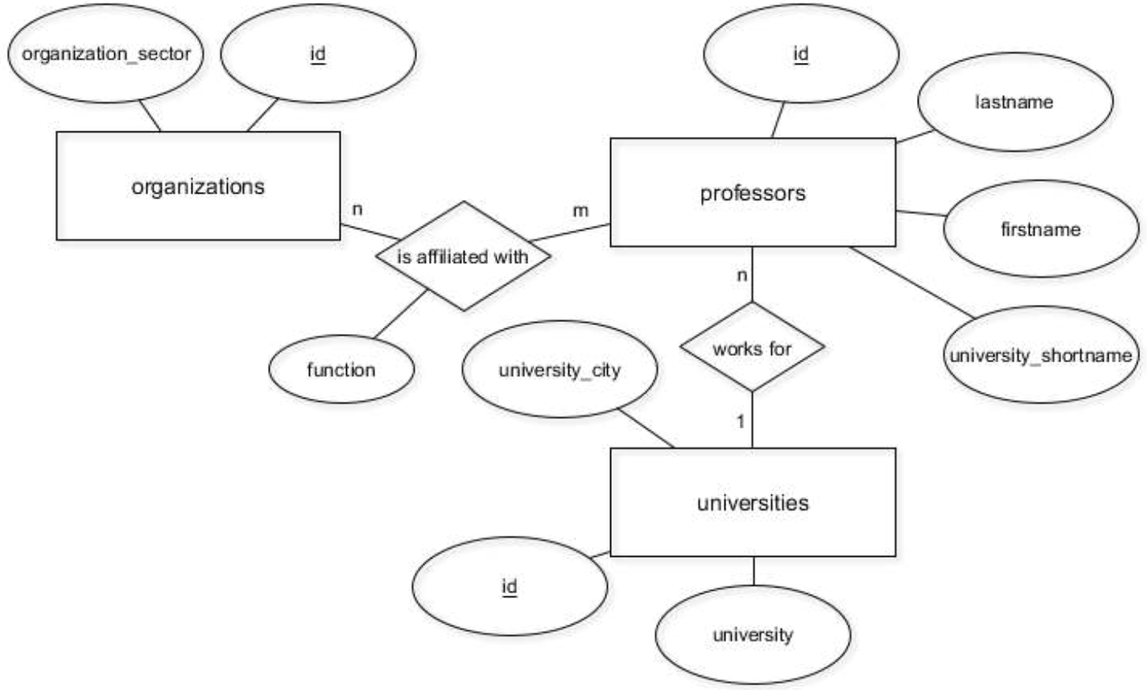
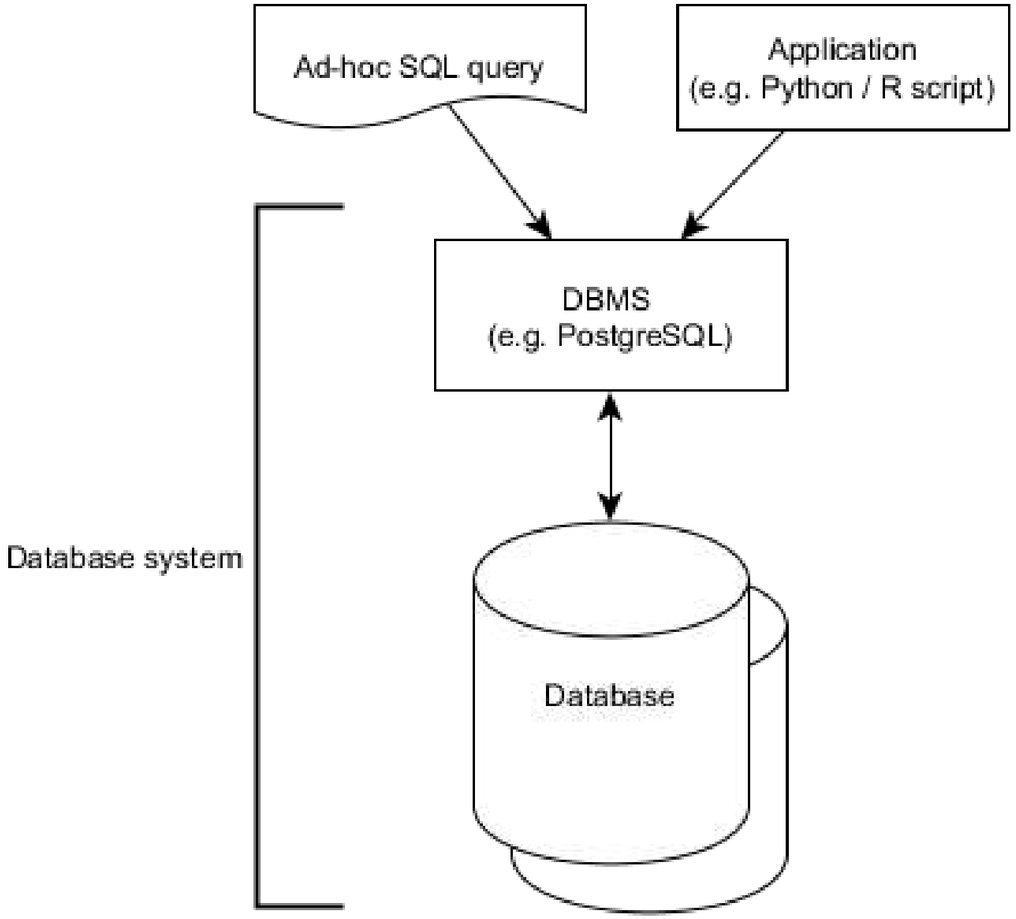
How you've transformed the database



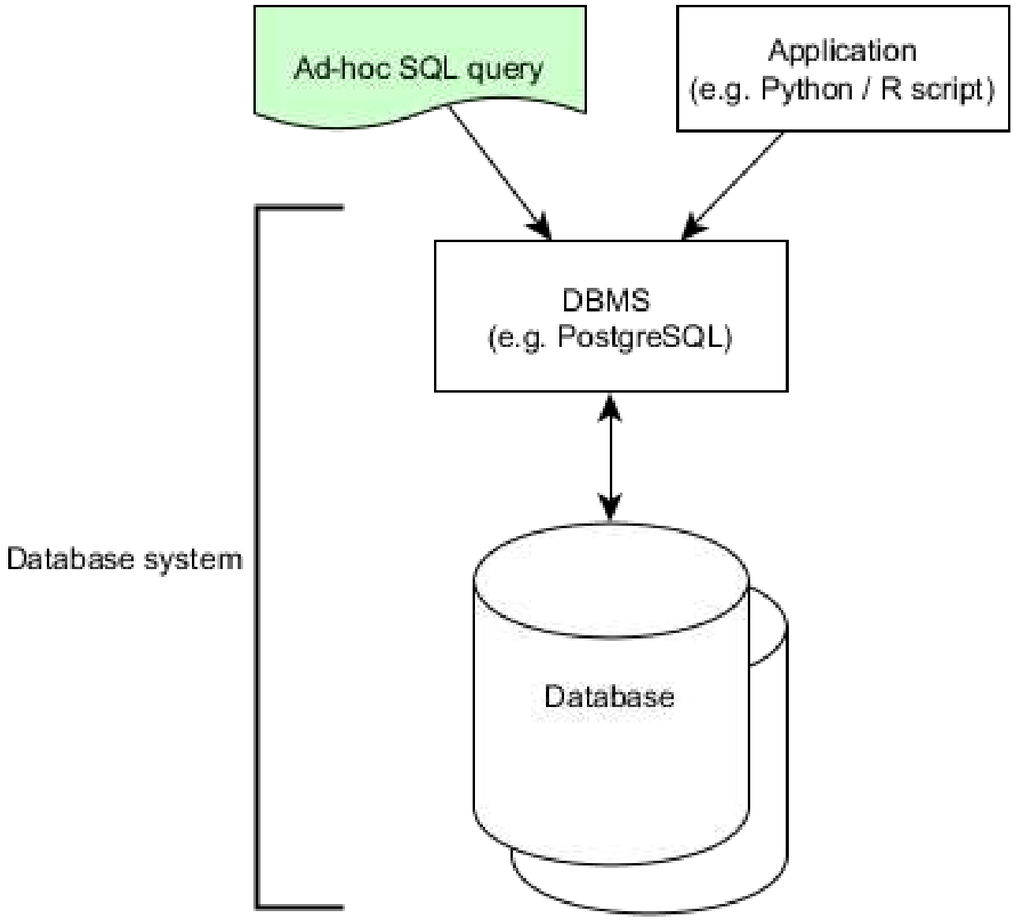


- Column data types
- Key constraints
- Relationships between tables

The database ecosystem



The database ecosystem



Thank you!

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